Hyperspectral Image Classification via Cross-Domain Few-Shot Learning With Kernel Triplet Loss

Ke-Kun Huang, *Member, IEEE*, Hao-Tian Yuan, Chuan-Xian Ren, *Member, IEEE*, Yue-En Hou, Jie-Li Duan, and Zhou Yang

***Abstract*— Limited labeled training samples constitute a challenge in hyperspectral image classification, with much research devoted to cross-domain adaptation, where the classes of the source and target domains are different. Current cross-domain few-shot learning (FSL) methods only use a small number of sample pairs to learn the discriminant features, which limits their performance. To address this problem, we propose a new framework for cross-domain FSL, considering all possible positive and negative pairs in a training batch and not just pairs between the support and query sets. Furthermore, we propose a new kernel triplet loss to characterize complex nonlinear relationships between samples and design appropriate feature extraction and discriminant networks. Specifically, the source and target data are simultaneously fed into the same feature extraction network, and then, the proposed kernel triplet loss on the embedding feature and the cross-entropy loss on the softmax output are used to learn discriminant features for both source and target data. Finally, an iterative adversarial strategy is employed to mitigate domain shifts between source and target data. The proposed method significantly outperforms state-of-the-art methods in experiments on four target datasets and one source dataset. The code is available at https://github.com/kkcocoon/CFSL-KT.**

***Index Terms*— Convolutional neural network (CNN), discriminant learning, generative adversarial networks, hyperspectral image (HSI) classification, triplet loss.**

I. Introduction

THE hyperspectral image (HSI) is an important type of remote sensing image. Different from ordinary RGB images, an HSI has hundreds of contiguous narrow spectral bands, effectively combining spatial and spectral information. The use of HSIs to classify spatial objects has many applications, such as mineral exploration, water pollution detection, natural disaster research, and biological threat detection [1].

A challenge in HSI classification is that there are too few labeled samples because it takes a lot of manpower and material resources to determine the class of each pixel. However, some experimental HSI data include enough labeled data [the source domain (SD)]. To classify new HSI data [the target domain (TD)], we can take advantage of discriminative information from the SD. In practical applications, the classes differ between the two domains, and we can obtain few labeled samples from the TD. This task is called cross-domain few-shot learning (FSL).

The cross-domain FSL task has been approached by a variety of methods [2], [3], [4], [5], [6], [7]. Liu et al. [2] first proposed deep FSL (DFSL) to address the small sample size problem of HSI classification, using a deep residual network based on a 3-D CNN to learn a metric space. Li et al. [3] first proposed a deep cross-domain FSL (DCFSL) to extend DFSL with different domains and distributions. DCFSL adopts a conditional adversarial domain adaptation strategy to learn a domain-adaptive feature embedded space, which can achieve domain distribution alignment. DCFSL simultaneously employs FSL in both source and target classes to learn transferable knowledge in the source classes and a discriminative embedding model in the target classes.

However, existing cross-domain FSL methods only use some sample pairs to learn the discriminant features, i.e., they select a subset of samples within each class to act as a support set and a subset of the remainder to act as a query set, and these are fed into a neural network with shared parameters. They calculate the Euclidean metric of the network output features of each support and query sample from the same class and train the network to minimize the metric. The key is that the network learns a metric space from the labeled data, and this can generalize to the classes of the target data, even those not seen in the source dataset. Dividing labeled samples into a support and query set to learn a metric space is called FSL.

FSL does not consider pairs in the query set or in the support set. For example, in Fig. 1(a), samples 1 and 4 are selected as the support set and other samples as the query set. For sample 1, as we can see, FSL will pull sample pairs (1, 2) and (1, 3) together (solid line in the figure) and push sample pairs

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Ke-Kun Huang is with the School of Mathematics, Jiaying University, Meizhou, Guangdong 514015, China (e-mail: kkcocoon@163.com).

Hao-Tian Yuan and Jie-Li Duan are with the College of Engineering, South China Agricultural University, Guangzhou 510642, China (e-mail: hr8yhtzb@outlook.com; duanjieli@scau.edu.cn).

Chuan-Xian Ren is with the School of Mathematics, Sun Yat-sen University, Guangzhou 510275, China (e-mail: rchuanx@mail.sysu.edu.cn).

Yue-En Hou is with the School of Computer Science, Jiaying University, Meizhou, Guangdong 514015, China (e-mail: houyueen@jyu.edu.cn).

Zhou Yang is with the College of Engineering, South China Agricultural University, Guangzhou 510642, China, and also with the School of Mechanical Engineering, Guangdong Ocean University, Zhanjiang 524088, China (e-mail: yangzhou@scau.edu.cn).

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(1, 5) and (1, 6) apart (dashed line). However, FSL neglects sample pairs (1, 4), (2, 3), (2, 5), (2, 6), (3, 5), (3, 6), and (5, 6). Specifically, FSL will not push samples 1 and 4 apart, so it loses some useful discriminant information in the training set, which limits performance.

To address this problem, we propose a new framework for cross-domain FSL, which computes a distance matrix for all possible positive and negative pairs in a training batch and then selects the hardest pairs to compute the loss. As shown in Fig. 1(b), the distance matrix is of size 6×6, which considers more information than the FSL metric. Because the hardest negative sample of sample 1 is sample 4, the proposed metric will push samples 1 and 4 apart. Similarly, because the hardest positive sample of sample 1 is sample 3, samples 1 and 3 will be pulled closer together. Other samples also push the hardest negative samples far from each other and pull the hardest positive samples together. Our proposed method can more effectively utilize the useful information in the training set. Furthermore, we propose a new kernel triplet loss to characterize complex nonlinear relationships between samples, and we design appropriate feature extraction and discriminant networks to better align the SD and TD. The proposed cross-domain FSL based on kernel triplet loss is called CFSL-KT.

CFSL-KT makes the following contributions.

1) We propose a framework for cross-domain FSL, considering all possible positive and negative pairs in a training batch, to fully extract all discriminant information.
2) We propose a new kernel triplet loss, which can map samples to a higher dimensional space for better classification, to characterize complex nonlinear relationships between samples.
3) We design an appropriate adversarial network to better align the SD and TD. The proposed method significantly improves classification accuracy compared with state-of-the-art HSI classification methods.

The remainder of this article is organized as follows. Section II introduces work related to HSI classification. Section III describes the proposed methods. Experimental results are given in section IV, and we provide the conclusion in Section V.

II. Related Work

We introduce work related to HSI classification. We present some traditional methods with handcrafted features and shallow classifiers, and state-of-the-art deep learning methods based on convolutional neural networks (CNNs) and domain adaptation for HSI classification.

A. Traditional Methods

Traditional HSI classification methods can be divided into two categories: handcraft feature extraction method and shallow classifier. The handcraft feature extraction methods extract features at different scales and directions, such as Gabor wavelet [8] and local binary pattern [9]. It is worth noting that mathematical morphology can extract spatial features more effectively [10]. Benediktsson et al. [11] proposed a method based on morphological profiles (MPs) applied to HSIs, where MP is a sequence of multiscale connected operators. Morphological attribute profile (MAPs), a generalization of MP, was proposed in [12], which uses morphological attribute filters based on the geodesic reconstruction. In [13], extended MAP (EMAP) was presented for the classification of HSIs. EMAP uses multiple morphological attributes to replace the fixed structure elements, which models the spatial information of HSI more accurately.

The shallow classifiers include support vector machines (SVMs) [14], sparse representation and classification (SRC) [15], Markov random field (MRF) [16], extreme learning machine (ELM) [17], and so on. For these traditional classification methods, the feature extraction step is separate from the classification step, and they cannot be directly connected end to end. Thus, the performance of the traditional classification method is not good enough for HSI classification.

B. CNN-Based Methods

CNNs have excellent image-classification capabilities and are used in end-to-end classification methods [18]. The functional units of convolutional layers are kernels that can extract the most relevant and enriched spatial and spectral features from a dataset through automated filtering by convolution [19]. Many types of CNN-based HSI classification methods have been proposed, such as 1-D CNN for spectral bands [20], the combination of 1-D and 2-D CNNs [21], spatial–spectral feature extraction CNN [22], 3-D CNN [23], [24], and residual and dense CNNS [25], [26]. Specifically, Roy et al. [27] proposed a hybrid spectral CNN (HybridSN) that combined 2-D and 3-D CNNs for HSI classification, where the 3-D CNN extracts the joint spatial–spectral feature from a stack of spectral bands, and a 2-D CNN on top learns more abstract-level spatial representations. Zhong et al. [24] proposed a spectral–spatial residual network (SSRN), whose residual blocks use identity mapping to connect every other 3-D convolutional layer. Huang et al. [28] proposed a discriminant Gabor ensemble filter (DGEF) for HSI classification, combining traditional Gabor filters and CNN to extract deep features for HSI with fewer trainable parameters.

CNN-based methods can classify HSIs well and have excellent feature extraction capabilities. However, training millions of parameters in a CNN requires a large number of labeled training samples, which are difficult to collect. By contrast, the human visual system can recognize new classes from a few labeled samples. To mimic the human learning style, FSL has received extensive attention [2]. Gao et al. [29] used metalearning to train a RelationNet model with a small number of samples. The model can fully learn transferable feature knowledge and relation comparison ability through metalearning. Jiang et al. [30] introduced a CNN-based pyramid structure to generate features in multiresolution and also learn a similar relation model to compare class prototypes and queries in multiscales. Sun et al. [31] proposed a metalearning method based on earth mover distance for HSI few-shot classification, which aims to find the optimal matching flows

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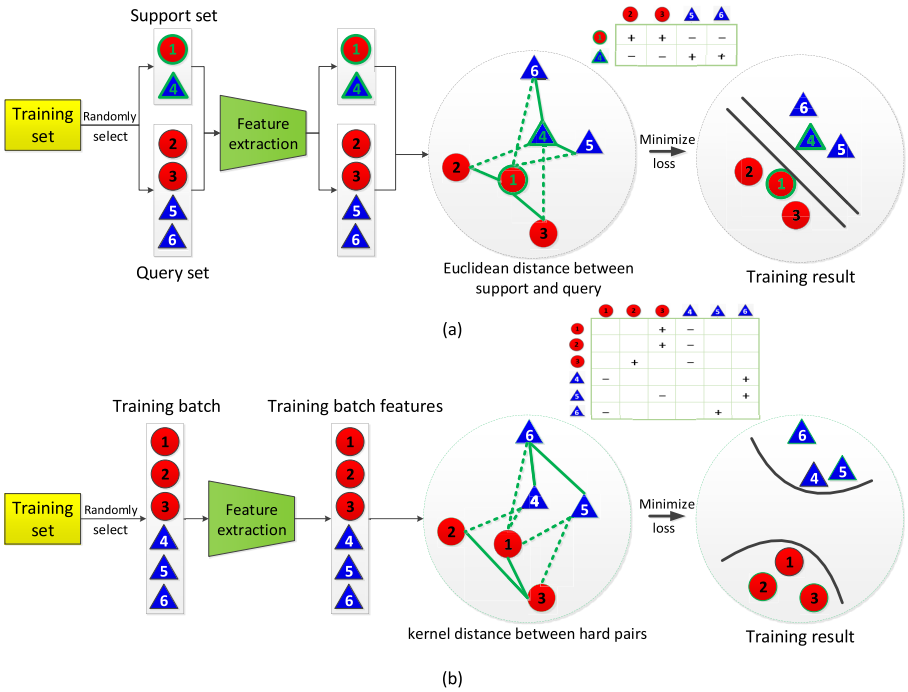


Fig. 1. Comparison of (a) FSL metric and (b) proposed metric. FSL only considers sample pairs between the support and query sets to learn a metric space, which does not consider pairs in the query or support set, where the distance matrix is of size 2×4. The proposed metric first computes a kernel matrix for all possible positive and negative pairs in a training batch, where the distance matrix is of size 6×6, and then selects the hardest pairs for each sample to compute the loss. For example, the hardest positive pair for sample 2 is (2, 3), and the hardest negative pair is (2, 4). The proposed metric can get a larger discriminative margin than the FSL metric in the training result.

between local representations that have the minimum matching cost.

C. Domain Adaption Methods

In practice, the SD and TD may come from different sensors and regions, whose data have different feature distributions. Therefore, the metric space obtained by an SD does not perform well on a TD when using CNN-based and FSL methods. Recent research tried to overcome this problem through domain adaption methods [32], [33], [34], [35], [36].

Unsupervised domain adaption methods are applied for HSI classification when the SD and TD are with the same classes. Deng et al. [37] proposed an unsupervised domain-adaptive method based on deep distance learning. Yu et al. [38] proposed an unsupervised domain-adaptive method based on dense compressed CNN. Tang et al. [39] used a dual-classifier adversarial-based approach to deal with samples at decision boundaries. The above methods employ generative adversarial networks to align the edge distributions of sample features in SD and TD. In addition, some researchers have used unsupervised learning for HSI classification by utilizing encoder and decoder architectures [40], [41].

Cross-domain FSL methods can be applied for HSI classification when the classes differ between the SD and TD [3]. Deng et al. [37] proposed a deep metric learning-based model for HSI classification [similarity-based deep metric model (SDMM)], which can learn the similarity between a pair of samples using a small number of labeled samples. The method can be applied to both same- and cross-domain HSI classification. Wang et al. [42] proposed heterogeneous FSL (HFSL) for HSI classification. This method used mini-Imagenet datasets to train the VGGNet and transfer its parameters to extract spatial features of HSI. Zhang et al. [4] proposed a graph neural network information propagation and graph alignment strategy (GiaCFSL), with an intradomain distribution block to generate the distribution graph and graph optimal transport for cross-domain alignment. Xi et al. [5] proposed an FSL framework for HSI classification with a class-covariance metric (CMFSL). Compared with the commonly used Euclidean metric, CMFSL uses the class-covariance metric to construct more flexible decision boundaries. Class labels are determined directly using Mahalanobis distance measurement rather than an extra classifier in the test phase. Fang et al. [43] proposed confident learning-based domain adaptation (CLDA) that combines domain adaptation and label denoising for cross-domain HSI classification. In addition, some researchers have used multisource data [44], contrastive learning [45], classwise attention [46], and feature integration [47] to learn discriminative features in SD and TD for HSI classification.

III. Proposed Method

We present our proposed method. We describe the proposed CFSL-KT framework, present the proposed kernel triplet loss, discuss the proposed losses for discriminative learning in SD and TD, and describe the domain alignment loss.

A. Framework

SD and TD refer to two different HSIs, where the source HSI has enough labeled samples, while the target one only has

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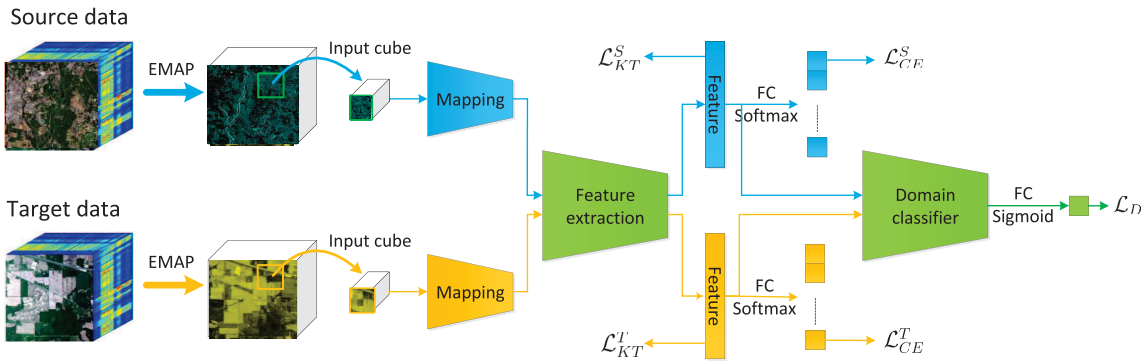


Fig. 2. Proposed CFSL-KT framework. The input sample is a cube centered on each pixel of EMAP features for the original HSI. Two mapping layers without share weights are used to map two domains to the same input dimension. Feature extraction network with shared parameters extracts embedding features of SD and TD. Kernel triplet loss and cross-entropy loss are used to learn discriminant features. Domain classifier aligns domain distribution.

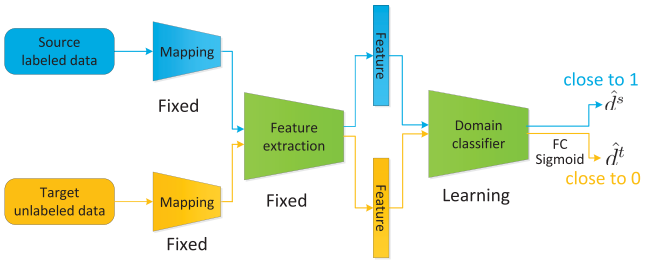


Fig. 3. Training domain classifier. The mapping layer and the feature extraction network are fixed, i.e., their parameters are not updated during training. We only train domain classifiers, which will distinguish samples from SD or TD. If a sample is from the SD, then the output value is expected to be as close to 1 as possible. Otherwise, it is expected to be closer to 0.

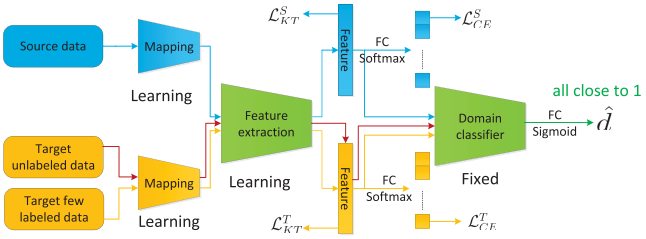


Fig. 4. Training feature extraction network. The mapping layer and the feature extractor, with parameters updated, are used to make extracted embedded features discriminative and domain-invariant. The outputs of the domain classifiers are all expected to be as close to 1 as possible.

few labeled samples, and we need to classify the unlabeled target samples.

Fig. 2 shows the proposed CFSL-KT framework. We first extract EMAP features for the original HSI and then extract a patch centered on each pixel to create a cube as an input sample for the network. Then, two mapping layers without shared weights and a feature extraction network with shared parameters are used to extract the embedding features of the SD and TD and to align them to a domain-invariant feature space.

EMAP feature can model the spatial information of HSI more accurately and can reduce the number of data channels. The feature extraction network with shared parameters may not work well enough for different domains, so the mapping layers without shared weights are used. We found that EMAP and the mapping layer can improve the performance of the experiment. Thus, we use the EMAP feature and mapping in the proposed method.

After that, kernel triplet loss on the embedding feature and cross-entropy loss on the softmax output are used to learn discriminant features. Kernel triplet loss considers all possible positive and negative pairs in a training batch, instead of only considering pairs between the support and query sets, which can characterize complex nonlinear relationships between samples. Finally, a conditional domain discriminator is utilized to mitigate domain shifts between domains.

The CFSL-KT framework has two processes, to alternately train a domain classifier and a feature extraction network.

Fig. 3 shows the process of training a domain classifier. The mapping layer and the feature extraction network are fixed without updating parameters, while the domain classifier is learnable. We only train the domain classifier to better distinguish the SD from the TD. The inputs of the domain classifier are the embedding features output from the feature extraction networks of SD and TD samples, and the domain classifier outputs a value between 0 and 1, indicating that the probability of the sample is from the SD or TD. A sample with an output value close to 1 is more likely from the SD. The loss $\mathcal{L}_D$, as defined in the following, makes the outputs of the domain classifiers for SD samples as close to 1 as possible and makes the outputs for TD samples closer to 0.

Fig. 4 shows the process of training the feature extraction network. The domain classifier is fixed, while the mapping layer and the feature extraction network are learnable. The embedding features of labeled samples from the SD and TD are used to calculate the losses, $\mathcal{L}_{\mathrm{KT}}^{S}$ and $\mathcal{L}_{\mathrm{KT}}^{T}$, which will be defined in the following, respectively, to learn the discriminant features. The softmax outputs of labeled samples from the SD and TD are used to calculate the losses, $\mathcal{L}_{\mathrm{CE}}^{S}$ and $\mathcal{L}_{\mathrm{CE}}^{T}$, respectively. The embedding features of unlabeled samples from the TD are also fed into the domain classifier to help domain alignment. It should be noted that we only train the feature extraction network so that the outputs of the domain classifiers are all expected to be as close to 1 as possible. Hence, the feature extraction network will fool the domain classifier so that it cannot distinguish whether the samples are from the SD or TD, i.e., the samples from the SD and TD are mapped to a domain-invariant embedding feature space.

The total loss for training the feature extraction network is

$$\mathcal{L}_{\mathcal{F}} = \lambda_K \left(\mathcal{L}_{\mathrm{KT}}^{S} + \mathcal{L}_{\mathrm{KT}}^{T}\right) + \mathcal{L}_{\mathrm{CE}}^{S} + \mathcal{L}_{\mathrm{CE}}^{T} + \lambda_D \mathcal{L}_D \tag{1}$$

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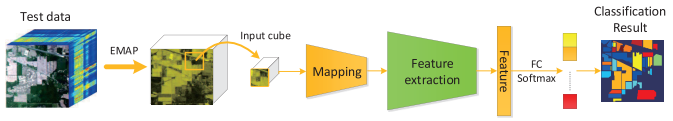


Fig. 5. Flowchart of classification in the testing phase. Softmax output of testing data is directly used for classification.

where λ_K and λ_D are weighted to balance the kernel triplet loss and the domain classifier loss, respectively. $\mathcal{L}_{\mathrm{KT}}^{S}$, $\mathcal{L}_{\mathrm{KT}}^{T}$, $\mathcal{L}_{\mathrm{CE}}^{S}$, and $\mathcal{L}_{\mathrm{CE}}^{T}$ are defined in the following. It is notable that the proposed losses $\mathcal{L}_{\mathrm{KT}}^{S}$ and $\mathcal{L}_{\mathrm{KT}}^{T}$ consider all the possible positive pairs and negative pairs in a training batch, instead of only considering the pairs between the support set and the query set, to fully extract all discriminant information.

Algorithm 1 CFSL-KT

Input: Source dataset S, target dataset T_b and T, learning rate η, loss weight λ_K and λ_D;
Output: Feature extraction network and domain classifier with parameters $\Theta_{\mathcal{F}}$ and $\Theta_{\mathcal{D}}$, respectively;

Network parameter initialization;
for *iterations* **do**
  Sample data from S, T_b and T:
  $\mathcal{S} = \{(\boldsymbol{x}_i^s, \boldsymbol{y}_i^s)\}$, $\mathcal{T}_b = \{(\boldsymbol{x}_i^b, \boldsymbol{y}_i^b)\}$ and $\mathcal{T} = \{\boldsymbol{x}_i^t\}$;
  % Training domain classifier
  Compute $\mathcal{L}_{\mathcal{D}}$ according to Eq. (11);
  Update model via backpropagation:
    $\Theta_{\mathcal{D}} \leftarrow \Theta_{\mathcal{D}} - \eta \nabla \mathcal{L}_{\mathcal{D}}(\Theta_{\mathcal{D}}; \mathcal{S}, \mathcal{T}, \mathcal{T}_b)$;
  % Training feature extraction network
  Compute $\mathcal{L}_{\mathcal{F}}$ according to Eq. (1), (7), (8), (9), (10), and (12);
  Update model via backpropagation:
    $\Theta_{\mathcal{F}} \leftarrow \Theta_{\mathcal{F}} - \eta \nabla \mathcal{L}_{\mathcal{F}}(\Theta_{\mathcal{F}}; \mathcal{S}, \mathcal{T}_b)$;
end

The training pipeline is presented in Algorithm 1. CFSL-KT alternately trains the domain classifier and the feature extraction network. In summary, $\mathcal{L}_{\mathcal{D}}$ aims to achieve a domain classifier with excellent domain distinguishing ability, and $\mathcal{L}_{\mathcal{F}}$ aims to let feature extraction network achieve discriminative and domain-invariant feature.

For the testing phase, we directly use the softmax output of the testing data for classification, whose flowchart is shown in Fig. 5.

B. Kernel Triplet Loss

The samples of HSI have dramatic intraclass diversity and interclass similarity, and it is not enough to use Euclidean distance to characterize the complex relationship between samples. Inspired by metric learning methods [48], [49] and kernel methods [50], [51], we propose kernel triplet loss, considering all possible pairs in a training batch and mapping the samples to a higher dimensional space for better classification, as shown in Fig. 6.

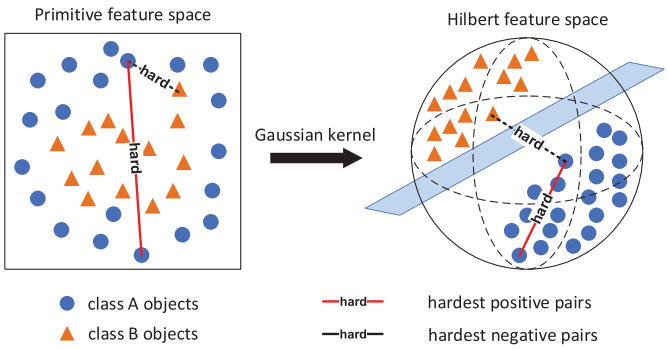


Fig. 6. Proposed kernel triplet hard loss. In a primitive feature space, the distance of the hardest positive pair (connect with the black dashed line) is much greater than that of the hardest negative pair (connect with the red solid line). The proposed kernel triplet hard loss can map samples to a high-dimensional feature space where samples can be more easily classified.

The kernel triplet loss is defined as

$$\mathcal{L}_{\mathrm{KT}} = \frac{1}{N}\sum_{i=1}^{N} h\left(\alpha + \max_{p,\boldsymbol{y}_p=\boldsymbol{y}_i} \mathcal{K}(\boldsymbol{x}_i, \boldsymbol{x}_p) - \min_{n,\boldsymbol{y}_n\neq\boldsymbol{y}_i} \mathcal{K}(\boldsymbol{x}_i, \boldsymbol{x}_n)\right) \tag{2}$$

where $\boldsymbol{x}_i$ is a sample and N is the number of samples. $\mathcal{K}(\boldsymbol{x}_i, \boldsymbol{x}_j)$ is a kernel distance, which is defined in (3). $(\boldsymbol{x}_i, \boldsymbol{x}_p)$ is a positive pair, and $(\boldsymbol{x}_i, \boldsymbol{x}_n)$ is a negative pair. $\max_{p,\boldsymbol{y}_p=\boldsymbol{y}_i} \mathcal{K}(\boldsymbol{x}_i, \boldsymbol{x}_p)$ means that only the hardest positive pair for $\boldsymbol{x}_i$ contributes to the loss, and $\min_{n,\boldsymbol{y}_n\neq\boldsymbol{y}_i} \mathcal{K}(\boldsymbol{x}_i, \boldsymbol{x}_n)$ means that only the hardest negative pair contributes. $h(x) = \max(0, x)$ is the hinge loss function, and α is a margin to filter trivial pairs.

The kernel distance function is defined as

$$\mathcal{K}(\boldsymbol{x}_i, \boldsymbol{x}_j) = \sum_{k=1}^{N_k}\left(1 - e^{-\frac{\|f(\boldsymbol{x}_i) - f(\boldsymbol{x}_j)\|^2}{\sigma_k}}\right) \tag{3}$$

where $f(\boldsymbol{x}_i)$ is the embedding feature output of the feature extraction network f for sample $\boldsymbol{x}_i$. $\|f(\boldsymbol{x}_i) - f(\boldsymbol{x}_j)\|^2$ is the Euclidean distance between $f(\boldsymbol{x}_i)$ and $f(\boldsymbol{x}_j)$. σ_k is a standard deviation, which is defined as

$$\sigma_k = \frac{\sum_{i=1}^{N}\sum_{j=1}^{N}\|f(\boldsymbol{x}_i) - f(\boldsymbol{x}_j)\|^2}{N^2 - N} \cdot 2^k \tag{4}$$

i.e., σ_k is the average distance multiplied by 2^k. We use five standard deviations with different values, i.e., $N_k = 5$. The function value ranges from 0 to N_k. The more similar the two samples, the closer the value to 0.

According to the proposed kernel triplet loss, for each sample $\boldsymbol{x}_i$, all samples from the same class act as candidates to construct positive pairs, and all samples from different classes act as candidates to construct negative pairs. Only the hardest of the candidate positive pairs is used to calculate the loss.

C. Discriminative Learning in SD and TD

There are two FSL-KTs in CFSL-KT, which are applied to the SD and TD, respectively. The labeled data from the SD are used to train the feature extraction network to extract discriminative features, and we believe that this ability is also

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applicable to the TD. Few labeled data from the TD are also used to train the feature extraction network.

Let $S = \{(\boldsymbol{x}_i^s, \boldsymbol{y}_i^s)|i = 1, 2, \ldots, N_s\}$ denote the labeled samples from the SD, $T_b = \{(\boldsymbol{x}_i^b, \boldsymbol{y}_i^b)|i = 1, 2, \ldots, N_b\}$ the few labeled samples from the TD, and $T = \{\boldsymbol{x}_i^t|i = 1, 2, \ldots, N_t\}$ the unlabeled samples from the TD, where $\boldsymbol{y}_i^s = [y_{i,1}^s, y_{i,2}^s, \ldots, y_{i,C_s}^s]$ is a one-hot vector indicating the true label for training sample $\boldsymbol{x}_i^s$, $\boldsymbol{y}_i^b = [y_{i,1}^b, y_{i,2}^b, \ldots, y_{i,C_t}^b]$ is a one-hot vector indicating the true label for training sample $\boldsymbol{x}_i^b$, C_s is the number of classes of the SD, C_t is the number of classes of the TD, and N_s, N_b, and N_t are the numbers of labeled source samples, labeled target samples, and unlabeled target samples, respectively.

To learn discriminative features, most deep learning-based methods use cross-entropy loss to train the network

$$\begin{aligned}\mathcal{L}_{\mathrm{CE}} &= -\frac{1}{N}\sum_{i=1}^{N} < \boldsymbol{y}_i, \log \hat{\boldsymbol{y}}_i > \\ &= -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c} \log \hat{y}_{i,c} \end{aligned} \tag{5}$$

where $\hat{\boldsymbol{y}}_i = [\hat{y}_{i,1}, \hat{y}_{i,2}, \ldots, \hat{y}_{i,C}]$ is the network softmax output for sample $\boldsymbol{x}_i$.

For binary classification, the derivative of cross-entropy $\mathcal{L}_{\mathrm{CE}}$ is

$$\begin{aligned}\frac{\partial \mathcal{L}_{\mathrm{CE}}}{\partial \hat{y}_i} &= -\frac{\partial\left(y_i \log\left(\hat{y}_i\right) + (1 - y_i)\log\left(1 - \hat{y}_i\right)\right)}{\partial \hat{y}_i} \\ &= -\frac{y_i}{\hat{y}_i} + \frac{1 - y_i}{1 - \hat{y}_i}. \end{aligned} \tag{6}$$

According to (6), the cross-entropy loss learns faster when the fitting error is larger. However, the training process is guided by the high-probability samples, i.e., the easier ones ($\hat{y}_i \to 0$ or $\hat{y}_i \to 1$). For HSI classification, many samples are difficult to classify, i.e., they have similar spectra but different labels, or vice versa. Solely using the cross-entropy loss is insufficient to learn discriminative features.

To address samples that are difficult to classify, besides the cross-entropy loss, we combine the proposed kernel triplet loss to learn more discriminative features in the SD and TD.

The cross-entropy loss on the SD is

$$\mathcal{L}_{\mathrm{CE}}^{S} = -\frac{1}{N_s}\sum_{i=1}^{N_s} < \boldsymbol{y}_i^s, \log \hat{\boldsymbol{y}}_i^s > \tag{7}$$

where $\hat{\boldsymbol{y}}_i^s$ is the softmax output for the embedding feature $f(\boldsymbol{x}_i^s)$ and N_s is the number of SD samples.

The kernel triplet loss in the SD is

$$\mathcal{L}_{\mathrm{KT}}^{S} = \frac{1}{N_s}\sum_{i=1}^{N_s} h\left(\alpha + \max_{p, \boldsymbol{y}_p^s = \boldsymbol{y}_i^s} \mathcal{K}\left(\boldsymbol{x}_i^s, \boldsymbol{x}_p^s\right) - \min_{n, \boldsymbol{y}_n^s \neq \boldsymbol{y}_i^s} \mathcal{K}\left(\boldsymbol{x}_i^s, \boldsymbol{x}_n^s\right)\right) \tag{8}$$

where $\boldsymbol{x}_i^s$ is a sample from the SD, $\boldsymbol{x}_i^s$ and $\boldsymbol{x}_p^s$ have the same class, $\boldsymbol{x}_i^s$ and $\boldsymbol{x}_n^s$ have different classes, and α is a margin to filter trivial pairs.

Similarly, the cross-entropy loss on the TD is

$$\mathcal{L}_{\mathrm{CE}}^{T} = -\frac{1}{N_b}\sum_{i=1}^{N_b} < \boldsymbol{y}_i^b, \log \hat{\boldsymbol{y}}_i^b > \tag{9}$$

where $\hat{\boldsymbol{y}}_i^b$ is the softmax output for the embedding feature $f(\boldsymbol{x}_i^b)$ and $\boldsymbol{x}_i^b$ is a labeled sample from the TD.

The kernel triplet loss in the TD is

$$\mathcal{L}_{\mathrm{KT}}^{T} = \frac{1}{N}\sum_{i=1}^{N_b} h\left(\alpha + \max_{p, \boldsymbol{y}_p^b = \boldsymbol{y}_i^b} \mathcal{K}\left(\boldsymbol{x}_i^b, \boldsymbol{x}_p^b\right) - \min_{n, \boldsymbol{y}_n^b \neq \boldsymbol{y}_i^b} \mathcal{K}\left(\boldsymbol{x}_i^b, \boldsymbol{x}_n^b\right)\right). \tag{10}$$

D. Domain Alignment

As shown in Figs. 3 and 4, we alternate training of the domain classifier and the feature extraction network.

The loss when training the domain classifier is

$$\begin{aligned}\mathcal{L}_D = &-\frac{1}{N_s}\sum_{i=1}^{N_s} \log \hat{d}_i^s - \frac{1}{N_b}\sum_{i=1}^{N_b} \log\left(1 - \hat{d}_i^b\right) \\ &- \frac{1}{N_t}\sum_{i=1}^{N_t} \log\left(1 - \hat{d}_i^t\right) \end{aligned} \tag{11}$$

where $\hat{d}_i^s$, $\hat{d}_i^b$, and $\hat{d}_i^t$ are the domain classifier outputs for $\boldsymbol{x}_i^s$, $\boldsymbol{x}_i^b$, and $\boldsymbol{x}_i^t$, respectively. $\boldsymbol{x}_i^s$ is a sample from the SD, and $\boldsymbol{x}_i^b$ and $\boldsymbol{x}_i^t$ are the samples from the TD. When the samples come from the SD, the loss $\mathcal{L}_D$ makes $\hat{d}_i^s$ close to 1. When the samples come from the TD, the loss $\mathcal{L}_D$ makes $\hat{d}_i^s$ and $\hat{d}_i^t$ close to 0. This means that the training process will update the parameters of the domain classifier to better distinguish the samples from the SD and TD.

The loss when training the feature extraction network is

$$\begin{aligned}\mathcal{L}_D = &-\frac{1}{N_s}\sum_{i=1}^{N_s} \log \hat{d}_i^s - \frac{1}{N_b}\sum_{i=1}^{N_b} \log \hat{d}_i^b \\ &- \frac{1}{N_t}\sum_{i=1}^{N_t} \log \hat{d}_i^t. \end{aligned} \tag{12}$$

Here, when the samples come from the TD, the loss $\mathcal{L}_D$ makes $\hat{d}_i^s$ and $\hat{d}_i^t$ close to 1. This means that the training process will update the parameters of the feature extraction network to better align the SD and TD.

E. Compared With Traditional FSL

The loss for the traditional FSL methods [2], [3] is defined as

$$\mathcal{L}_{\mathrm{FSL}} = -\frac{1}{N_q}\sum_{q=1}^{N_q}\sum_{k=1}^{C} y_{q,k} \log \frac{e^{-\|f(\boldsymbol{x}_q) - \boldsymbol{c}_k\|_2}}{\sum_{j=1}^{C} e^{-\|f(\boldsymbol{x}_q) - \boldsymbol{c}_j\|_2}} \tag{13}$$

where $\boldsymbol{c}_k = (1/N_k)\sum_{y_{i,k}=1} f(\boldsymbol{x}_i)$ is the mean embedding features of the kth class in the support set and C is the number of classes. $\boldsymbol{x}_q$ is the qth query sample, and $y_{q,k}$ is the corresponding one-hot class label. N_q is the number of the query samples.

Suppose that we select a total of $(K + Q) \times C$ samples (K samples for each class as the support set and Q samples

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for each class as the query set) to create a training episode. For example, $K = 5$, $Q = 10$, and $C = 5$. Then, the number of sample pairs in $\mathcal{L}_{\mathrm{FSL}}$ is only $Q \times C = 50$. For the proposed method, the number of sample pairs in $\mathcal{L}_{\mathrm{KT}}$ is $((K+Q)\times C)\times((K+Q)\times C) = 62500$, which is much more than that in $\mathcal{L}_{\mathrm{FSL}}$. Thus, the proposed method considers more information than the FSL metric and can get a larger discriminative margin than the FSL metric in the training result. Furthermore, we combine the proposed kernel triplet loss and the cross-entropy loss to learn more discriminative features.

F. Compared With Our Previous Work

In our previous work [10], an improved triplet loss [convolutional neural network with improved triplet loss (CNN-IT)] was proposed for HSI classification, which is defined as

$$\mathcal{L}_{\mathrm{IT}} = \sum_{c=1}^{C}\sum_{k=1}^{K_c} h\left[\alpha + \frac{1}{N_{c,k}^{p}} \sum_{\substack{j=1\cdots K_c \\ D > D_{c,k}^{p}}} \|f(\boldsymbol{x}_{c,k}) - f(\boldsymbol{x}_{c,j})\|_D^2 - \frac{1}{N_{c,k}^{n}} \sum_{\substack{i=1\cdots C \\ j=1\cdots K_i \\ i\neq c \\ D < D_{c,k}^{n}}} \|f(\boldsymbol{x}_{c,k}) - f(\boldsymbol{x}_{i,j})\|_D^2\right] \tag{14}$$

where $\boldsymbol{x}_{c,k}$ means the kth sample in class c and the item $(1/N_{c,k}^{p})\sum_{\substack{j=1,\ldots,K_c \\ D > D_{c,k}^{p}}} \|f(\boldsymbol{x}_{c,k}) - f(\boldsymbol{x}_{c,j})\|_D^2$ indicates the mean of the distances between $\boldsymbol{x}_{c,k}$ and its top several hardest positive samples.

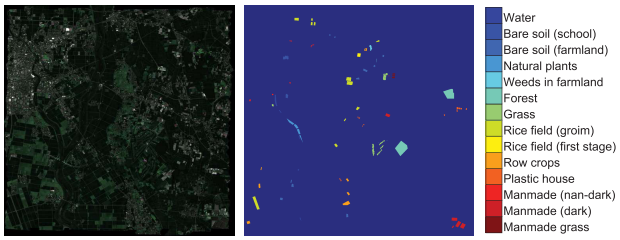
Compared (2) with (14), we can find that the proposed CFSL-KT and CNN-IT [10] are different as follows.

1) CNN-IT improves the triplet hard loss by selecting the top several hardest pairs instead of only selecting the hardest one. However, the proposed kernel triplet loss first maps samples to a high-dimensional feature space where samples can be more easily classified, which can characterize complex nonlinear relationships between samples, and then selects the hardest one to calculate the loss.
2) CNN-IT is not a domain adaption method, which only proposed a new loss function and is not designed for FSL, while the proposed CFSL-KT proposed a new cross-domain FSL framework, which can take advantage of discriminative information from the SD.

Thus, the proposed method is essentially different from CNN-IT. We will see in the experiments that the proposed method achieves better performance than CNN-IT [10].

IV. Experiments

We implemented the proposed method using PyTorch and the Python language. The experiments were conducted on a single GPU of an NVIDIA Titan X in the Python language. We introduce the experimental data, describe the parameter settings, and discuss the comparison with state-of-the-art methods along with ablation experiments.



(a) (b) (c)

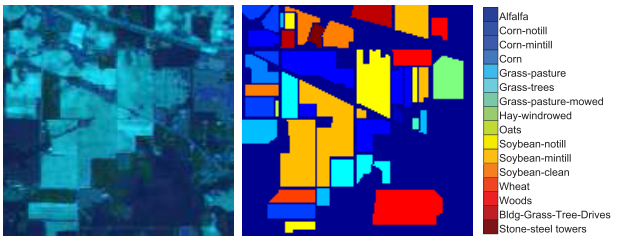
Fig. 7. Chikusei dataset (source). (a) False-color image. (b) Ground truth. (c) Class names.

A. Experimental Data

The performance of CFSL-KT was assessed on five public HSI datasets: Chikusei, University of Pavia (UP), Salinas, Indian Pines (IP), and Houston. Chikusei was selected as the SD dataset and the others as TD datasets.

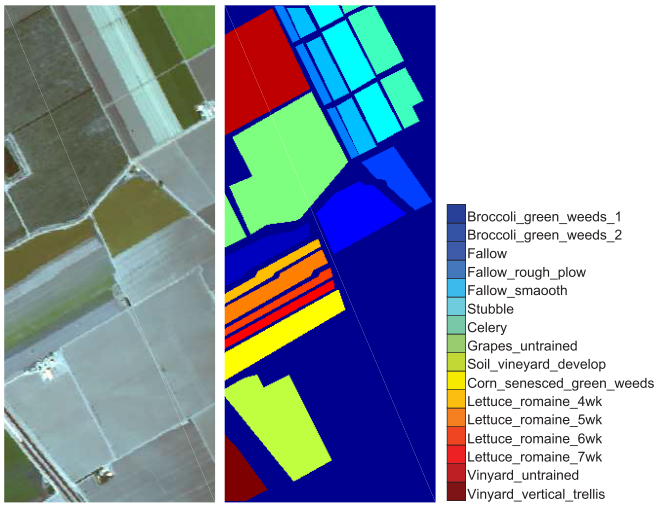
1) The Chikusei dataset was collected by Hyperspectral Visible/Near-Infrared Cameras (Hyperspec-VNIR-C) in Chikusei, Ibaraki, Japan, on July 29, 2014. It consists of 128 spectral bands with a size of 2517 × 2335 and a spatial resolution of 2.5 m, which ranges from 363 to 1018 nm. The ground truth contains 19 classes, including urban and rural areas. We selected as the SD dataset the 14 classes that had more than 1000 samples. The false-color image and corresponding ground-truth map are shown in Fig. 7.
2) The IP dataset was imaged by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) in 1992 on a farm in Indiana, USA. It contains 145×145 pixels. It consists of 220 spectral bands, from which 200 remain after removing bad bands. The spatial resolution is 20 m. There are 16 types of crops, including alfalfa, corn, and wheat. False-color images and the corresponding ground-truth map are shown in Fig. 8.
3) The Salinas data were collected by the AVIRIS sensor in Salinas Valley, California, USA, including 224 bands, with a size of 512 × 217 pixels. The spatial resolution is 3.7 m. After deleting invalid bands, 200 bands were used in experiments. The false-color image and corresponding ground-truth map are shown in Fig. 9.
4) The UP dataset is a portion of the hyperspectral data imaged by the Reflective Optics Spectrographic Imaging System (ROSIS-03) in 2003 in Pavia, Italy. Its 103 spectral bands range from 430 to 860 nm. Its spatial resolution is 1.3 m per pixel, and the size of the data is 610 × 340. It contains nine classes of ground truth, including trees and asphalt roads. The false-color image and corresponding ground-truth map are shown in Fig. 10.
5) The Houston was collected by ITRES CASI-1500 in Houston, TX, USA, in 2013. It consists of 144 spectral bands with a size of 349 × 1905 and a spatial resolution of 2.5 m, which ranges from 364 to 1046 nm. It contains 15 classes of ground truth, including road, soil, tree, and highway. The false-color image and corresponding ground-truth map are shown in Fig. 11.

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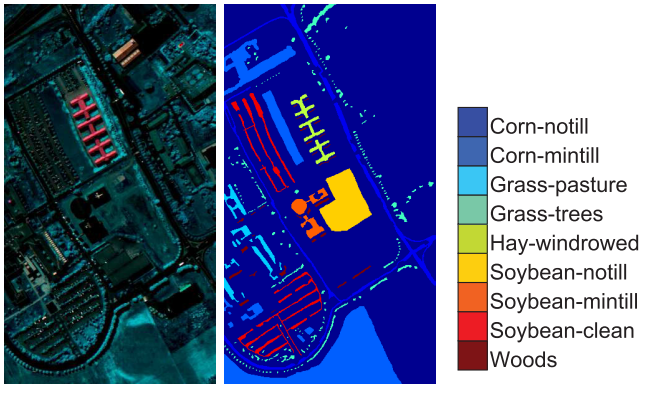
(a) (b) (c)

Fig. 8. IP dataset (target). (a) False-color image. (b) Ground truth. (c) Class names.



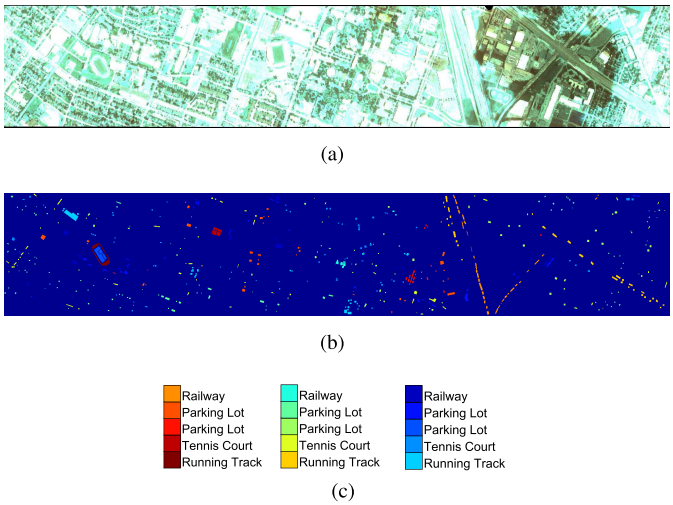
(a) (b) (c)

Fig. 9. Salinas dataset (target). (a) False-color image. (b) Ground truth. (c) Class names.



(a) (b) (c)

Fig. 10. UP dataset (target). (a) False-color image. (b) Ground truth. (c) Class names.



(a) (b) (c)

Fig. 11. Houston dataset (target). (a) False-color image. (b) Ground truth. (c) Class names.

TABLE I

PARAMETERS OF MAPPING LAYER AND FEATURE EXTRACTION NETWORK

Layer	Output Shape	Filter Size	Padding
Input	9 9 ch	N/A	N
Mapping	9 9 128	1 1 128	N
Convolution 1	7 7 128	3 3 128	N
Batch Norm	7 7 128	N/A	N
ReLU	7 7 128	N/A	N
Convolution 2	5 5 128	3 3 128	N
Batch Norm	5 5 128	N/A	N
ReLU	5 5 128	N/A	N
Convolution 3	5 5 64	1 1 64	N
Batch Norm	5 5 64	N/A	N
ReLU	5 5 64	N/A	N
Flatten	1600	N/A	N
Fully Connect (output)	256	N/A	N

TABLE II

PARAMETERS OF DOMAIN CLASSIFIER

Layer name	Output Shape	Factor
Input	1 1 256	N/A
Fully Connect 1	1 1 1024	N/A
ReLU	1 1 1024	N/A
Dropout	1 1 1024	0.5
Fully Connect 2	1 1 1024	N/A
ReLU	1 1 1024	N/A
Dropout	1 1 1024	0.5
Fully Connect 3	1 1 1024	N/A
ReLU	1 1 1024	N/A
Dropout	1 1 1024	0.5
Fully Connect 4	1 1 1024	N/A
ReLU	1 1 1024	N/A
Dropout	1 1 1024	0.5
Fully Connect (output)	1 1 1	N/A

B. Parameter Setting

For the proposed CFSL-KT method, 9×9 neighborhoods were selected as the spatial size of the input cubes. The parameters of the mapping layer, feature extraction network, and domain classifier are shown in Tables I and II.

The first step of the proposed method is to extract EMAP features [13]. We consider two attributes of EMAP: 1) the area of the regions $\lambda_a = [100, 500, 1000, 5000]$ and 2) the standard deviation of the gray-level values of the pixels in the regions $\lambda_s = [20, 30, 40, 50]$.

To compute the discriminative loss, we randomly selected 128 samples as a training batch and used all the samples if there were not enough. The proposed method was trained with an SGD optimizer, with a momentum of 0.99 and a weight decay of 0.0001. We set a base learning rate of 0.001. The number of training iterations was set to 2000. We randomly

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TABLE III
CLASSIFICATION RESULTS OF STATE-OF-THE-ART METHODS ON THE IP DATASET

Class	Train(#)	Test(#)	HybridSN	SSRN	DGEF	CNN-IT	DFSL	SDMM	CLDA	DCFSL	HFSL	CMFSL	GiaCFSL	**CFSL-KT**
1	5	41	95.36	98.78	98.54	96.11	97.44	98.05	93.26	90.73	98.25	95.61	92.44	98.57
2	5	1423	24.62	34.98	47.60	41.26	38.34	49.69	66.34	46.57	51.15	49.81	43.90	44.86
3	5	825	31.19	52.84	58.37	66.06	43.35	48.36	36.72	48.53	63.13	58.80	45.81	72.30
4	5	232	47.89	77.76	90.34	65.58	68.45	81.72	91.39	73.97	85.37	82.97	76.21	88.45
5	5	478	58.12	76.86	78.68	71.26	70.21	82.59	86.34	75.71	70.29	75.02	74.00	74.94
6	5	725	87.52	92.63	87.52	96.44	76.38	95.92	94.21	84.65	79.13	86.10	88.44	96.80
7	5	23	99.56	100.0	100.0	96.50	99.77	100.0	93.21	99.13	97.73	99.13	98.26	100.0
8	5	473	97.91	80.95	99.43	98.33	75.67	78.73	90.98	85.96	99.32	86.15	86.74	99.24
9	5	15	100.0	100.0	100.0	100.0	99.00	100.0	97.00	99.33	95.71	100.0	98.00	100.0
10	5	967	49.70	62.58	63.19	75.4	47.90	77.19	70.03	64.73	67.22	63.26	60.91	78.82
11	5	2450	41.49	57.28	56.82	65.82	57.80	41.00	21.01	60.51	68.66	56.65	63.40	73.99
12	5	588	28.98	54.35	54.86	49.89	38.13	40.30	21.21	43.15	78.38	51.31	45.53	61.87
13	5	200	96.45	98.35	97.65	99.94	98.04	99.90	88.24	97.35	98.34	99.45	98.85	100.0
14	5	1260	69.55	84.78	91.83	83.73	83.08	84.92	80.45	82.51	80.66	80.61	79.74	91.71
15	5	381	59.00	73.99	87.01	92.81	62.86	71.55	85.85	59.58	89.37	73.70	72.44	83.66
16	5	88	93.29	99.54	97.16	99.37	99.94	99.32	99.78	98.96	88.97	99.32	99.09	99.77
Overall Accuracy (%)			51.28 ±4.74	64.68 ±4.68	69.06 ±3.58	70.76 ±3.02	58.94 ±1.85	63.47 ±5.62	58.95 ±4.18	64.92 ±2.71	72.05 ±4.04	66.26 ±2.24	64.68 ±2.74	**76.29 ±1.26**
Average Accuracy (%)			67.54 ±3.25	77.85 ±3.36	81.81 ±1.97	81.16 ±1.87	72.27 ±0.84	78.08 ±2.48	76.01 ±6.38	75.70 ±1.77	81.98 ±3.07	78.62 ±0.96	75.98 ±2.09	**85.36 ±1.21**
Kappa Coefficient			47.36 ±5.22	62.07 ±5.45	67.17 ±0.04	67.18 ±3.29	53.01 ±1.90	59.19 ±6.08	54.43 ±4.62	60.50 ±2.90	68.54 ± 4.48	62.09 ±2.34	60.11 ±2.97	**73.41 ±1.39**
Training time (minutes)			1	1	4	6	35	3	4	37	83	69	106	16
Test time (seconds)			3	4	5	2	4	13	1	3	4	3	2	3

TABLE IV
CLASSIFICATION RESULTS OF STATE-OF-THE-ART METHODS ON THE SALINAS DATASET

Class	Train(#)	Test(#)	HybridSN	SSRN	DGEF	CNN-IT	DFSL	SDMM	CLDA	DCFSL	HFSL	CMFSL	GiaCFSL	**CFSL-KT**
1	5	2004	100.0	98.21	99.25	97.73	94.86	98.92	99.71	98.94	96.22	99.08	99.49	97.51
2	5	3721	96.99	84.25	99.64	99.25	98.68	99.77	93.67	99.81	95.26	99.69	98.36	99.38
3	5	1971	93.85	93.13	98.96	99.71	93.65	93.02	56.18	91.54	95.17	92.71	87.78	99.85
4	5	1389	90.21	98.94	98.45	98.66	99.39	99.18	99.84	99.50	98.86	99.35	98.87	99.64
5	5	2673	90.96	94.88	90.15	98.26	90.60	97.96	97.26	91.58	97.27	93.47	86.89	98.10
6	5	3954	97.25	99.33	97.19	99.46	97.64	99.35	100.0	98.96	99.48	99.18	98.51	99.44
7	5	3574	98.22	99.85	99.51	100.0	99.33	99.94	99.84	99.01	99.67	99.12	97.89	100.0
8	5	11266	62.77	54.53	70.98	81.08	78.34	74.42	89.09	74.16	69.98	72.75	74.19	84.67
9	5	6198	99.33	99.39	99.28	99.85	90.19	99.46	99.43	99.17	99.76	99.82	98.19	99.98
10	5	3273	90.32	87.03	89.80	92.82	61.84	88.60	97.68	83.60	90.97	85.92	79.35	92.80
11	5	1063	94.01	95.34	99.56	98.96	96.45	98.61	98.34	97.69	97.20	97.67	95.76	99.81
12	5	1922	86.03	99.91	96.42	99.05	93.11	99.83	98.23	99.67	97.80	99.89	97.97	100.0
13	5	911	97.28	99.59	97.33	99.78	99.50	98.22	99.83	99.41	97.22	99.05	99.03	99.67
14	5	1065	97.75	96.51	94.70	96.20	97.53	95.53	89.68	98.42	96.21	97.71	97.05	99.72
15	5	7263	83.73	76.90	92.06	91.76	77.46	78.35	48.73	76.30	66.63	76.73	76.09	93.31
16	5	1802	89.69	93.40	98.91	100.0	85.98	90.45	97.84	91.22	95.85	91.66	88.61	100.0
Overall Accuracy (%)			86.87 ±4.04	84.43 ±4.12	91.00 ±3.07	94.06 ±1.72	87.67 ±1.30	90.05 ±2.16	87.75 ±1.65	89.02 ±1.99	87.42 ±3.38	89.38 ±2.01	87.48 ±2.27	**95.13 ±0.93**
Average Accuracy (%)			91.77 ±1.29	91.95 ±1.36	95.14 ±1.54	97.04 ±0.94	90.91 ±1.44	94.48 ±0.54	91.59 ±2.20	93.69 ±1.07	93.35 ±1.67	93.99 ±1.10	92.13 ±1.37	**97.69 ±0.52**
Kappa Coefficient			85.96 ±4.33	83.32 ±4.54	90.23 ±0.03	93.40 ±1.91	86.02 ±1.42	88.95 ±2.40	86.45 ±1.82	87.81 ±2.18	86.03 ±3.74	87.96 ±2.35	86.49 ±2.49	**94.60 ±1.03**
Training time (minutes)			1	3	6	15	56	10	17	40	113	71	112	17
Test time (seconds)			7	18	16	12	15	67	2	11	19	15	13	16

selected 200 samples for each class in the SD to learn the discriminative information and only selected five samples for each class in the TD to train the model if not specified. We set the margin $\alpha = 2$, the weight for kernel triplet loss $\lambda_K = 5$, and the weight for domain classifier loss $\lambda_D = 0.01$ for all datasets.

For more stable performance, we first used only TD samples to train the feature extraction network for 500 iterations and then added SD samples to train it for another 500 iterations. We then trained the domain classifier and feature extraction network alternately.

If not specified, five samples were randomly selected in the target classes for training. To ensure reliable experimental results, ten independent runs were executed for each method and dataset. The mean values of the overall accuracy (OA), average accuracy (AA) for each class, and kappa

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TABLE V
CLASSIFICATION RESULTS OF STATE-OF-THE-ART METHODS ON THE PAVIA UNIVERSITY DATASET

Class	Train(#)	Test(#)	HybridSN	SSRN	DGEF	CNN-IT	DFSL	SDMM	CLDA	DCFSL	HFSL	CMFSL	GiaCFSL	**CFSL-KT**
1	5	6626	30.57	62.36	63.14	97.75	69.05	82.83	80.80	74.48	80.52	79.71	79.90	98.79
2	5	18544	76.90	52.74	63.40	85.89	85.45	73.06	70.75	85.12	77.49	84.65	87.06	86.36
3	5	2094	80.75	43.04	74.78	83.18	56.83	85.04	80.82	67.39	84.31	65.13	61.53	86.71
4	5	3059	73.80	88.92	54.08	96.01	89.52	89.63	98.26	90.84	91.91	93.77	93.74	93.95
5	5	1340	98.70	98.67	97.72	99.93	99.29	99.98	99.99	99.12	99.63	99.53	99.18	99.25
6	5	5024	75.90	83.71	93.89	93.71	70.50	70.67	89.59	72.43	69.88	75.47	75.80	96.92
7	5	1325	95.44	87.01	97.42	99.65	71.36	98.42	81.03	79.18	92.77	76.82	76.17	99.92
8	5	3680	44.07	75.28	70.11	76.57	58.29	81.37	73022	60.51	93.37	79.38	66.57	73.78
9	5	942	60.78	99.00	67.78	97.39	97.02	99.96	99.48	97.62	99.90	98.18	98.29	94.42
Overall Accuracy (%)			67.64 ±7.87	65.45 ±5.67	69.63 ±4.90	89.56 ±3.52	78.23 ±1.04	79.01 ±4.72	79.08 ±3.92	79.93 ±4.64	81.46 ±4.58	83.28 ±3.66	82.38 ±1.95	**89.92 ±2.50**
Average Accuracy (%)			70.77 ±3.47	76.75 ±3.54	75.81 ±2.88	92.23 ±2.15	77.48 ±2.52	86.77 ±3.33	86.00 ±4.07	80.74 ±1.17	87.75 ±3.11	83.63 ±1.66	82.03 ±2.18	**92.22 ±1.93**
Kappa Coefficient			62.56 ±9.62	59.83 ±6.63	66.53 ±0.06	86.57 ±4.33	70.02 ±1.46	73.29 ±5.77	73.87 ±4.62	74.09 ±5.24	76.27 ±5.52	77.47 ±3.80	77.09 ±2.37	**87.03 ±3.11**
Training time (minutes)			1	2	4	5	48	9	23	35	61	43	69	15
Test time (seconds)			5	12	10	9	14	56	2	8	26	15	9	17

TABLE VI
CLASSIFICATION RESULTS OF STATE-OF-THE-ART METHODS ON THE HOUSTON DATASET

Class	Train(#)	Test(#)	HybridSN	SSRN	DGEF	CNN-IT	DFSL	SDMM	CLDA	DCFSL	HFSL	CMFSL	GiaCFSL	**CFSL-KT**
1	5	1246	72.30	73.00	81.06	82.41	83.60	94.18	89.74	89.20	89.61	89.74	88.80	85.12
2	5	1249	86.27	83.98	56.36	78.80	78.10	91.43	86.34	82.35	79.43	87.47	82.19	84.96
3	5	692	87.76	92.83	97.77	98.86	99.86	99.87	97.16	95.85	99.65	95.78	94.78	99.86
4	5	1239	62.88	80.95	60.05	77.14	81.20	94.84	91.46	91.71	90.72	92.28	89.90	83.31
5	5	1237	91.53	94.44	97.82	93.49	95.73	93.34	95.19	97.40	91.09	97.38	94.32	98.63
6	5	320	82.75	77.91	88.84	90.81	93.03	91.20	86.74	78.56	90.97	79.93	73.72	94.06
7	5	1263	30.72	67.28	46.90	74.04	67.01	74.32	69.51	64.18	86.93	73.76	63.33	81.73
8	5	1239	46.49	25.47	48.03	53.83	46.00	50.77	38.75	45.14	44.02	48.04	42.66	56.29
9	5	1247	28.32	81.09	60.53	67.76	71.12	76.20	77.68	65.95	68.67	70.12	59.17	73.52
10	5	1222	57.93	52.55	66.07	93.46	80.89	69.28	56.40	65.21	68.66	69.35	63.98	91.79
11	5	1230	48.75	49.67	71.55	84.76	83.23	73.98	62.46	64.11	68.76	66.73	62.50	86.99
12	5	1228	77.54	25.03	63.65	68.84	54.39	59.81	48.69	68.00	62.84	70.94	72.52	66.10
13	5	464	69.16	81.47	78.28	53.99	60.21	76.18	88.23	84.35	68.38	95.60	88.47	71.49
14	5	423	91.32	99.27	100.0	99.43	99.77	99.88	98.18	91.80	96.47	96.00	93.31	100.0
15	5	655	91.53	98.76	99.76	99.79	99.24	99.97	99.20	96.93	96.77	97.72	92.41	99.70
Overall Accuracy (%)			64.50 ±2.98	68.23 ±4.21	70.08 ±2.62	79.58 ±2.85	76.60 ±2.36	78.56 ±1.18	75.64 ±2.02	76.39 ±1.46	77.94 ±2.38	79.63 ±2.60	75.05 ±1.62	**82.63 ±1.68**
Average Accuracy (%)			68.35 ±2.81	72.25 ±4.16	74.45 ±2.40	81.16 ±2.62	79.13 ±2.51	83.06 ±1.62	79.02 ±1.64	78.72 ±1.25	80.20 ±2.27	82.06 ±1.40	77.47 ±1.69	**84.35 ±1.70**
Kappa Coefficient			61.51 ±3.15	65.44 ±4.75	68.12 ±3.32	77.92 ±3.08	74.73 ±2.54	79.23 ±1.88	73.71 ±2.16	74.49 ±1.57	76.16 ±2.56	77.99 ±2.21	73.05 ±1.74	**81.23 ±1.80**
Training time (minutes)			3	3	5	22	46	11	25	36	79	49	74	25
Test time (seconds)			10	50	18	18	22	65	5	14	20	21	15	20

coefficient were utilized to quantitatively estimate different methods.

C. Comparison of State-of-the-Art Methods

To validate the performance of the proposed method, we compared it with some other methods. HybridSN [27], SSRN [24], DGEF [28], and CNN-IT [10] are state-of-the-art deep learning-based methods. DFSL [2], SDMM [37], HFSL [42], DCFSL [3], CMFSL [5], GiaCFSL [4], and CLDA [43] are deep domain-adaptive and FSL-based algorithms.

We downloaded the source code for each compared method and set the parameters optimally. For traditional deep learning models (HybridSN, SSRN, DGEF, and CNN-IT), which only consider samples for training and testing in the same domain, only five samples per class were randomly selected for training, and the rest were used for testing. For domain-adaptive methods (DFSL, SDMM, DCFSL, CMFSL, GiaCFSL, and CFSL-KT), we randomly selected 200 samples in each class of the SD for learning transferable knowledge and five samples in each class of the TD for training. For HFSL, we used the mini-Imagenet dataset to train a VGGNet and transfer its parameters to a new VGGNet. For the unsupervised learning method CLDA, we clustered the samples in the TD into two regions, randomly selected 200 samples from each class of one region for training, and used all samples from the other region for testing.

1) Comparison of Classification Accuracy: Tables III–VI show the OA, AA, and kappa coefficient of different methods

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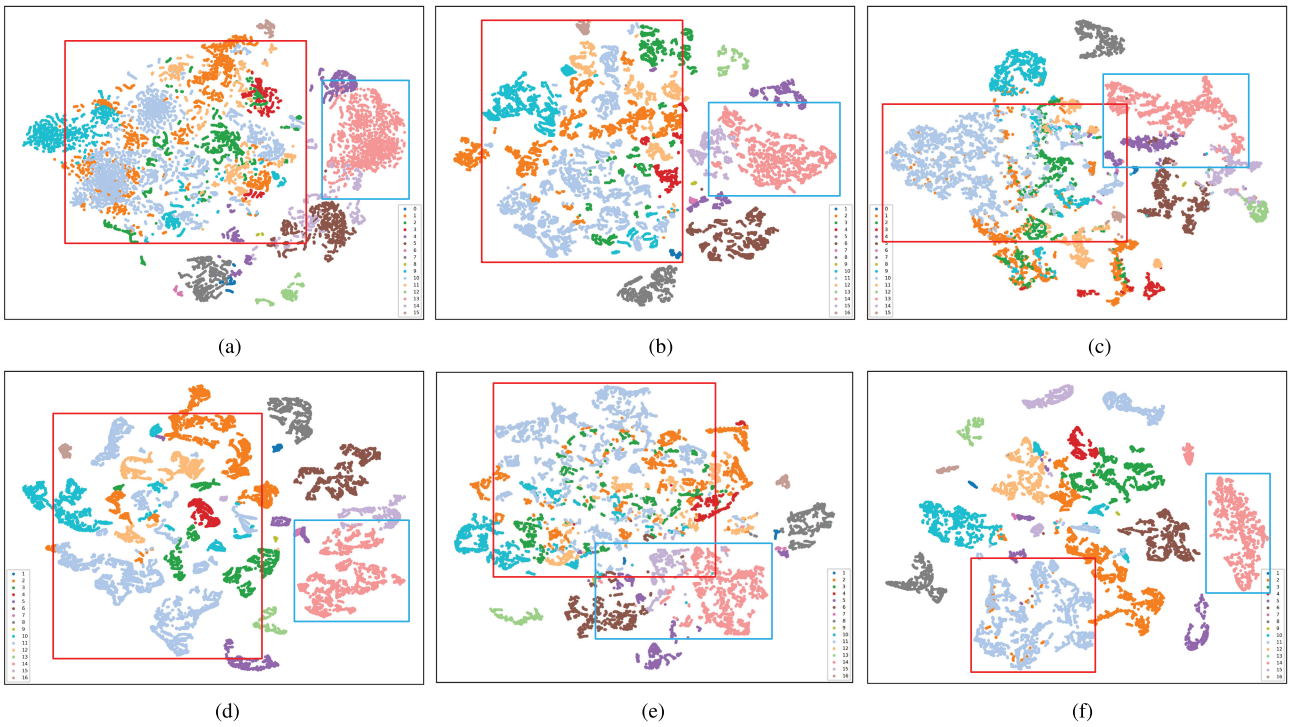


Fig. 12. t-SNE visualization of the embedding features learned by different cross-domain FSL methods on the IP dataset. Here, different colors represent different categories. It can be found that the proposed CFSL-KT can extract better discriminative features than other compared methods. (a) SDMM. (b) DCFSL. (c) HFSL. (d) CMFSL. (e) GiaCFSL. (f) CFSL-KT.

on the IP, Salinas, UP, and Houston datasets, respectively. From the results, we can observe the following.

1) HybridSN and SSRN have lower accuracy than other methods for most cases because they only use five TD samples for training, with no transferable knowledge from the SD.
2) DGEF achieves high performance on IP and Salinas datasets because it extracts Gabor features that do not need training, which is suitable for FSL. However, the performance of DGEF is lower than the cross-domain methods on PU and Houston datasets because the Gabor feature is not good for urban images.
3) CNN-IT achieves good results on four datasets, primarily due to its improved triplet hard loss function. This function effectively brings samples from the same class closer together while pushing apart samples from different classes. However, compared with the proposed method, CNN-IT exhibits a lower accuracy, especially on the IP dataset. This is because the proposed kernel triplet loss maps samples to a high-dimensional feature space where samples can be more easily classified. Moreover, CNN-IT is not a domain adaption method, while the proposed CFSL-KT can take advantage of discriminative information from the SD.
4) SDMM is better than HybridSN and SSRN on the Salinas, UP, and Houston datasets because it uses SD samples to learn transferable knowledge and uses a deep metric learning-based feature embedding model, which can complete tasks for cross-domain HSI classifications.
5) Cross-domain FSL methods, such as DCFSL, HFSL, CMFSL, and GiaCFSL, also outperform HybridSN and SSRN on four datasets because the FSL method constructs a metatask learning mode and can more effectively learn transferable knowledge with a few labeled samples.
6) The proposed method obtains the highest OA on all datasets. It improves a mean OA of 12.19%, 6.48%, 12.13%, and 7.88% for IP, Salinas, UP, and Houston datasets, respectively, compared with all methods. The AA and kappa coefficient of the proposed method are also significantly better than those of other methods. Compared with DFSL, the proposed method improves OA, with 17.35%, 7.64%, 11.69%, and 8.78% for IP, Salinas, UP, and Houston datasets, respectively. This is because the proposed framework for cross-domain FSL can overcome the shortage of traditional FSL methods, and the proposed new kernel triplet loss can map samples to a higher dimensional space for better classification.

2) Comparison of Feature Visualization: To analyze the quality of the learned embedding features, we employed t-SNE [52] to visualize the 2-D features generated by the different cross-domain FSL methods on the IP dataset, as illustrated in Fig. 12. It is obvious that the embedding features produced by the proposed CFSL-KT are more discriminative compared to the other methods. For example, the features of class 14 (highlighted in the blue box) produced by CFSL-KT are farther away from other classes than those produced by other methods, and the features of class 11 (highlighted in the red box) generated by CFSL-KT are more concentrated than those generated by other methods. Consequently, our proposed model demonstrates its ability to yield better classification

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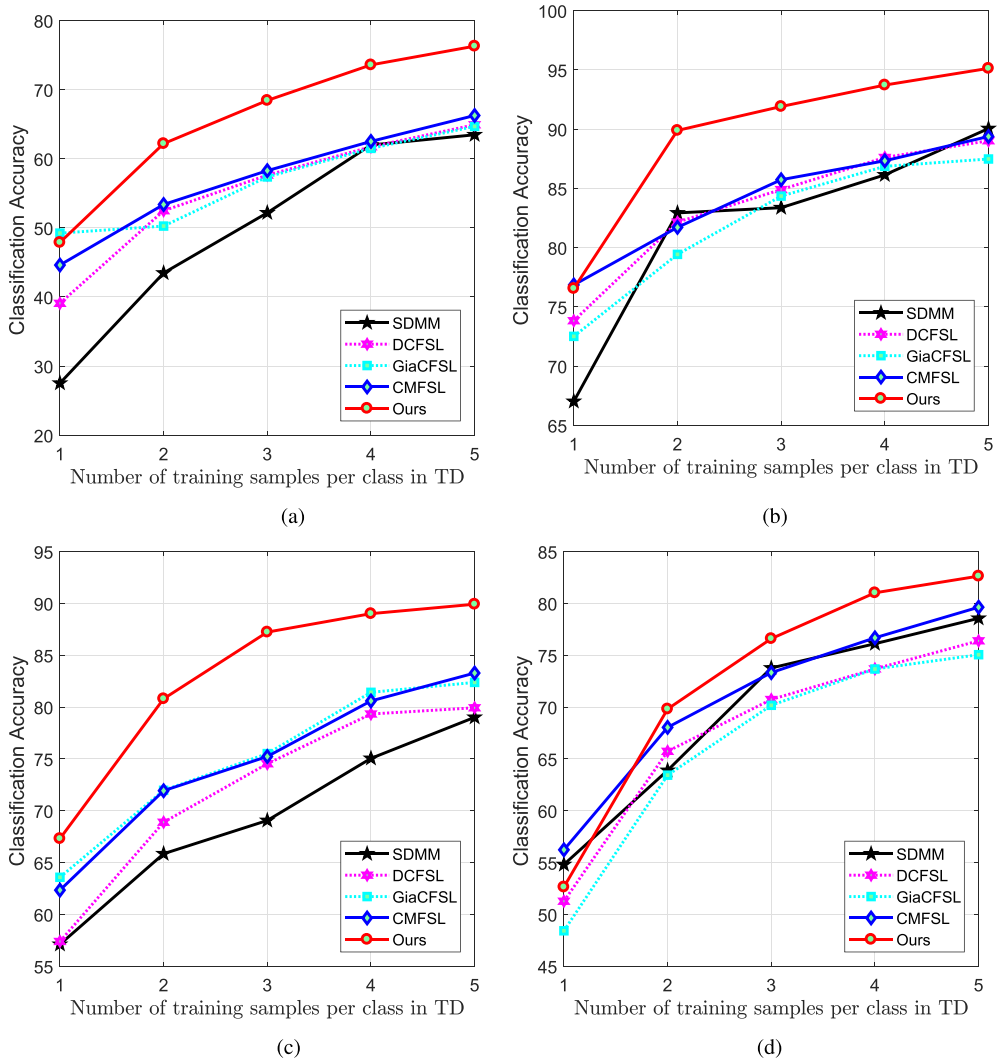


Fig. 13. Classification results (%) on four datasets with different numbers of training samples per class in TD. (a) IP. (b) Salinas. (c) PU. (d) Houston.

results. As shown in Table III, for class 11, the classification accuracies of SDMM, DCFSL, HFSL, CMFSL, and GiaCFSL are 41%, 60.51%, 68.66%, 56.65%, and 63.40%, respectively. In contrast, the proposed CFSL-KT achieves a significantly higher classification accuracy of 73.99%. For class 14, the classification accuracies of DCFSL, CMFSL, and GiaCFSL are 84.92%, 82.51%, 80.66%, 80.61%, and 79.74%, respectively, while the proposed CFSL-KT attains an impressive classification accuracy of 91.71%. These results demonstrate that the proposed CFSL-KT can extract better discriminative features than other compared methods.

3) Comparison of Training and Testing Times: The last row of Tables III–VI shows the training time in minutes and the testing time in seconds for different methods. All the methods were implemented in Python language. Experiments were conducted on an Intel Xeon processor (3.30 GHz), with 32 GB of memory, and an Nvidia TITAN Xp 12-GB graphics card. The testing time is the whole time for all of the testing samples. We can find the following.

1) The domain-adaptation methods take more training time than nondomain-adaptation methods because it requires a considerable amount of computing resources for domain adaptation to learn the transferable knowledge.
2) The training process of the proposed method takes less than 25 mins, which is faster than most of the domain-adaptation methods. For example, CMFSL and GiaCFSL take a few hours, and DCFSL takes more than half an hour.
3) The testing process of the proposed method is as fast as that of other methods, which only takes a few seconds for all the testing samples.

4) Comparison When Using Different Numbers of Labeled Samples From TD: We selected 15 labeled TD samples per class for training. Fig. 13 shows the classification accuracy of the domain-adaptation methods on the IP, Salinas, UP, and Houston datasets, respectively, from which we can observe the following.

1) When only using one TD sample, all methods have low classification accuracy, and the performance of CFSL-KT is not the best. This is because a good discriminant metric space cannot be learned using only one TD sample.

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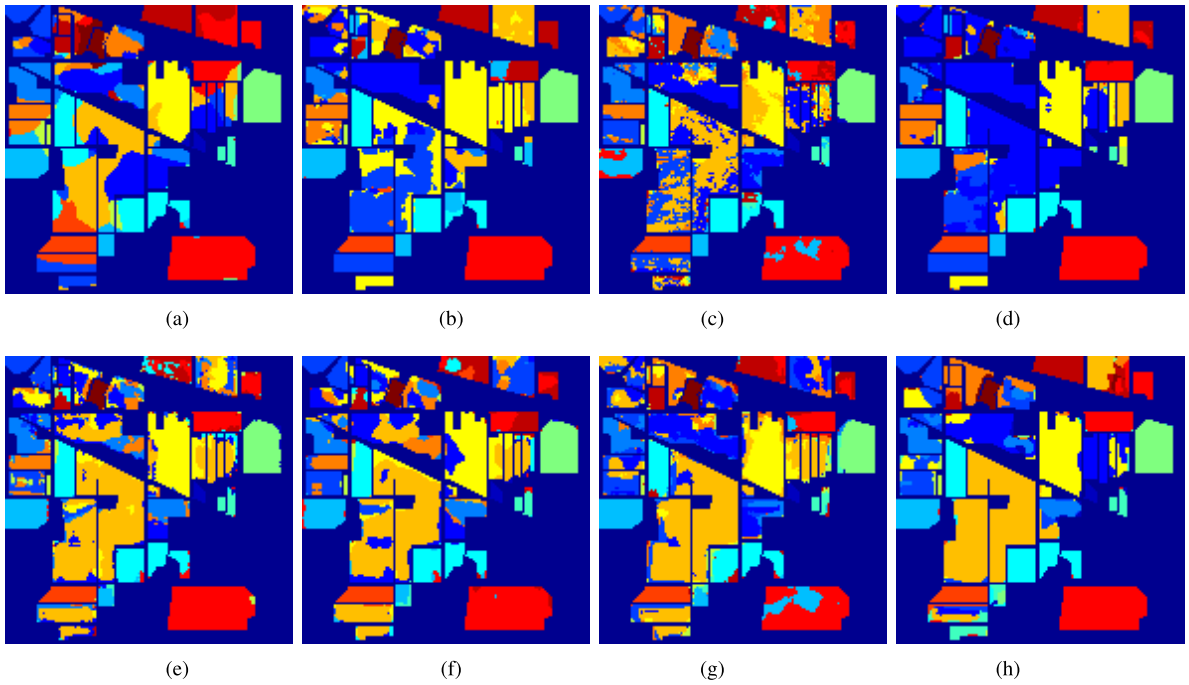


Fig. 14. Classification maps of methods using five training samples per class on the IP dataset. (a) DGEF (69.06%). (b) DFSL (58.94%). (c) SDMM (63.47%). (d) CLDA (58.95%). (e) DCFSL (64.92%). (f) CMFS (66.26%). (g) GiaCFSL (64.68%). (h) **CFSL-KT (76.29%).**

2) The proposed method significantly outperforms the others when two or more TD samples are used for training because the proposed kernel triplet loss can bring samples from the same class close to each other and push apart samples from different classes.

5) Comparison of Classification Maps: Figs. 14–16 show the classification maps of different methods using five TD samples per class on the IP, Salinas, and UP datasets, respectively. It can be easily found that many regions of the classification maps achieved by the proposed method are more accurate than those of other methods.

Fig. 17 shows the training loss and validation accuracy versus iterations for the proposed method. Here, iteration denotes the use of a batch of 128 samples to train the network. We can find that the loss and validation accuracy converge at about 1500 iterations. The loss suddenly increases at 500 iterations because we add TD samples for training at this time. The validation accuracy is stable after enough training iterations, which indicates that the proposed method does not suffer from overfitting.

D. Ablation Experiments

To explore the contribution of each part of the proposed method, we performed ablation experiments.

1) Contribution of Each Component of the Proposed Method: We performed ablation experiments for four cases.

1) *Without EMAP Preprocessing:* Because the network is designed especially for some handcraft features, not for raw HSI data, we use Gabor features [28] instead in this case.
2) *Without SD:* The network is only trained with TD samples, instead of both SD and TD samples.
3) *Without CE Loss:* Only kernel triplet loss is used for training. In the test phase, the nearest neighbor method is used to classify unlabeled data of TD.
4) *Without Mapping Layer:* The input of the feature extraction network is not performed by the mapping layer.
5) *Without KT Loss:* Only cross-entropy loss is used for training, instead of both cross-entropy and kernel triplet loss.
6) *Without Hilbert:* The distance between samples is not mapped to Hilbert space when computing triplet loss, i.e., $\mathcal{K}(\boldsymbol{x}_i, \boldsymbol{x}_j) = \|f(\boldsymbol{x}_i) - f(\boldsymbol{x}_j)\|^2$.
7) Proposed method.

For the ablation experiments, we only changed one component at a time, and the other settings remained the same.

Table VII shows the contribution of each component of the proposed method. From the results, we can observe the following.

1) When there is no EMAP, the performance dramatically decreases. EMAP feature can model the spatial information of HSI more accurately and can reduce the number of data channels. For the FSL problem, we cannot learn good convolutional features with few training samples. Thus, the learning system based on the handcraft EMAP feature can improve the performance in this scenario.
2) When there are no mapping layers, the performance slightly decreases because the feature extraction network with shared parameters may not work well enough for different domains, so the mapping layers without shared weights are used.
3) When we trained the network without SD samples, the classification accuracies on the IP, UP, Salinas, and Houston datasets were 75.93%, 88.69%, 95.10%, and 82.21%, respectively, which are lower than for the

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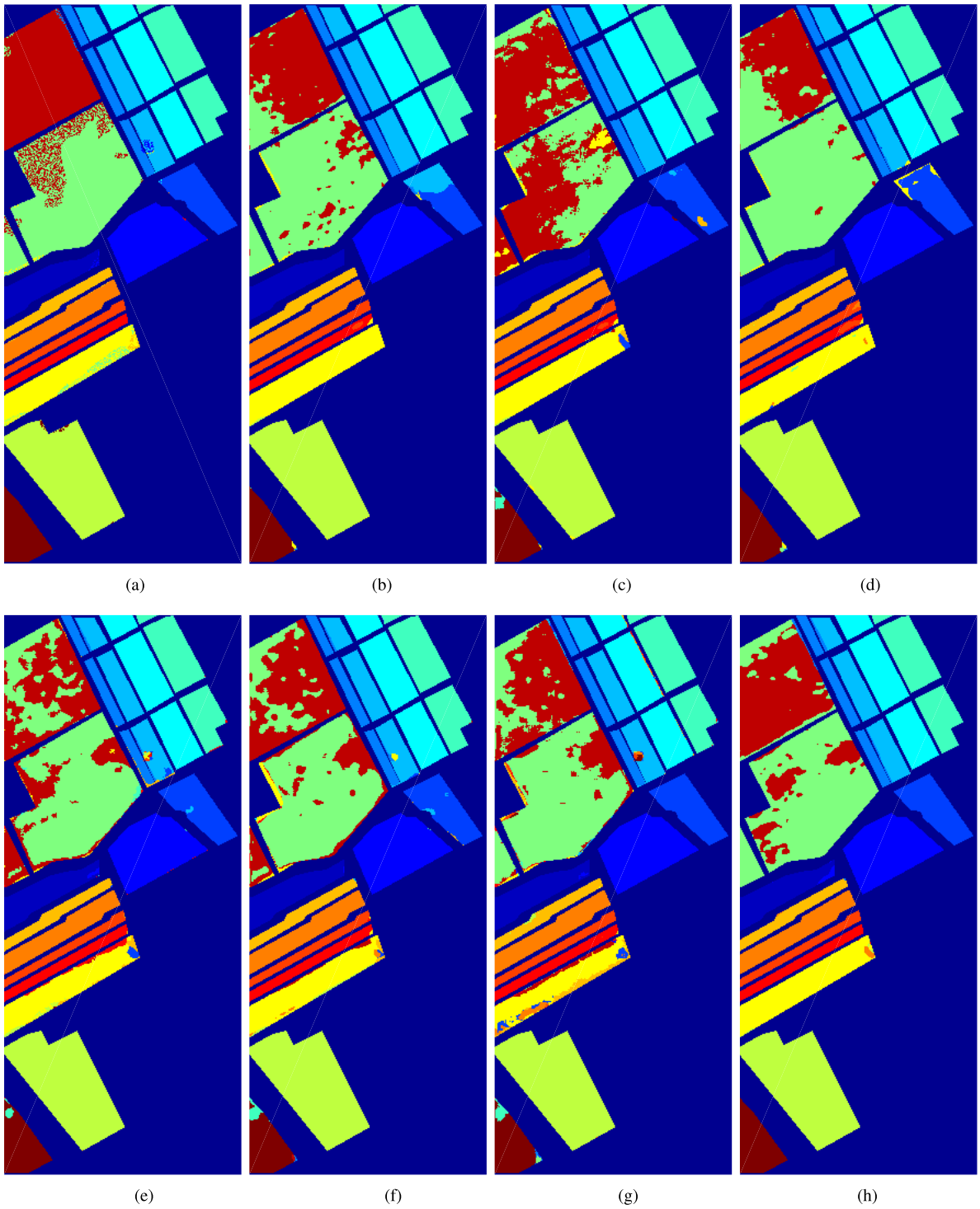


Fig. 15. Classification maps of methods using five training samples per class on the Salinas dataset. (a) DGEF (91.00%). (b) DFSL (87.67%). (c) SDMM (90.05%). (d) CLDA (87.75%). (e) DCFSL (89.02%). (f) CMFS (89.38%). (g) GiaCFSL (87.48%). (h) **CFSL-KT (95.13%).**

proposed method. This is because it does not utilize discriminative feature information from SD samples in this case. However, both cross-entropy and kernel triplet losses were used to train the network, so the classification accuracies were not dramatically reduced.

4) When we only trained with kernel triplet loss, without cross-entropy loss, the classification accuracies on the IP, UP, Salinas, and Houston datasets were 75.93%, 88.69%, 95.10%, and 78.41%, respectively, which are lower than for the proposed method.
5) When we only trained with cross-entropy loss, without kernel triplet loss, the classification accuracies on the IP, UP, Salinas, and Houston datasets were 71.34%, 86.04%, 90.89%, and 80.04%, respectively, which dramatically decreased. This demonstrates that the proposed kernel triplet loss can learn good discriminative features

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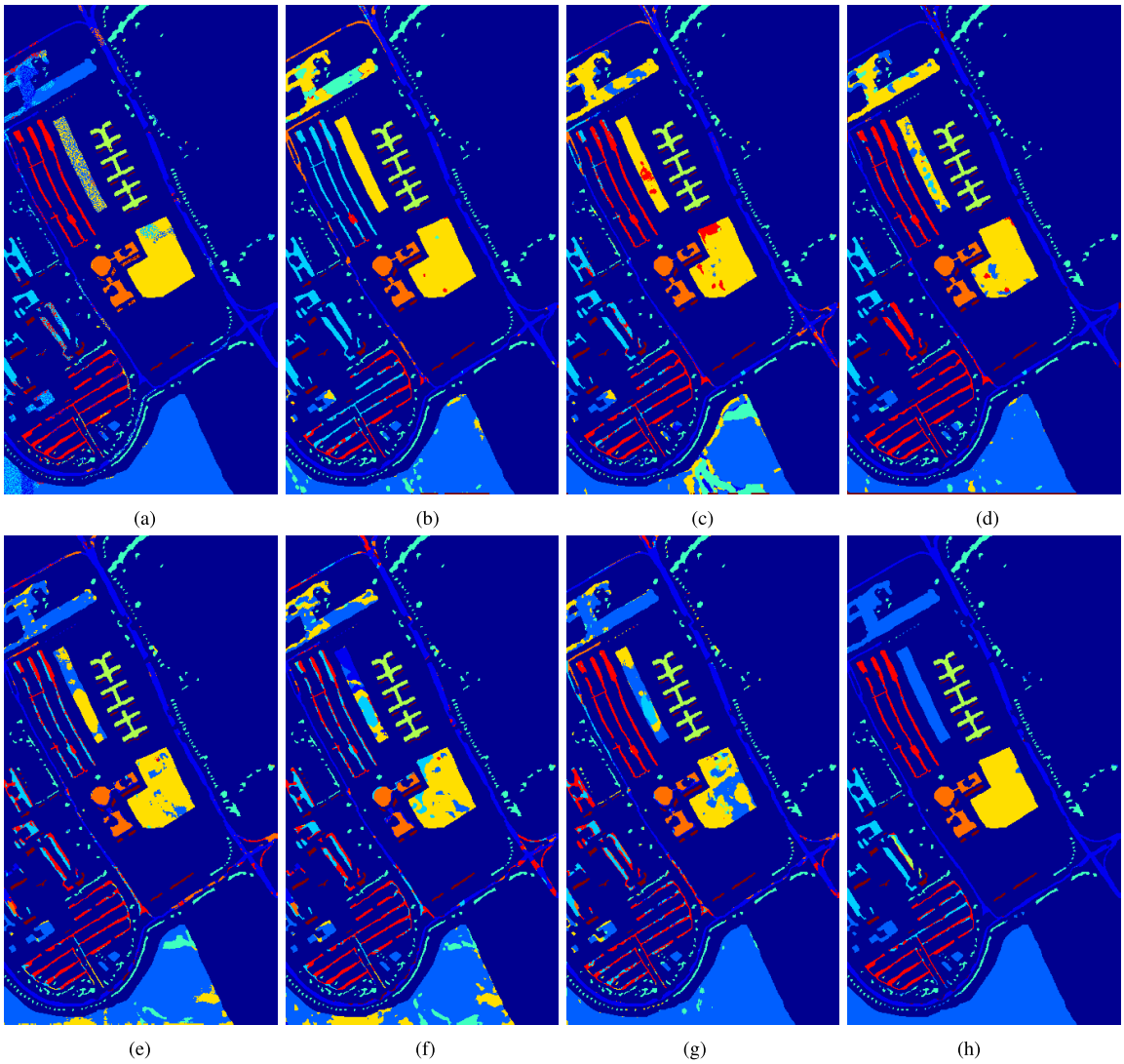


Fig. 16. Classification maps of methods using five training samples per class on the PU dataset. (a) DGEF (69.63%). (b) DFSL (78.23%). (c) SDMM (79.01%). (d) CLDA (79.08%). (e) DCFSL (79.93%). (f) CMFS (83.28%). (g) GiaCFSL (82.38%). (h) **CFSL-KT (89.92%).**

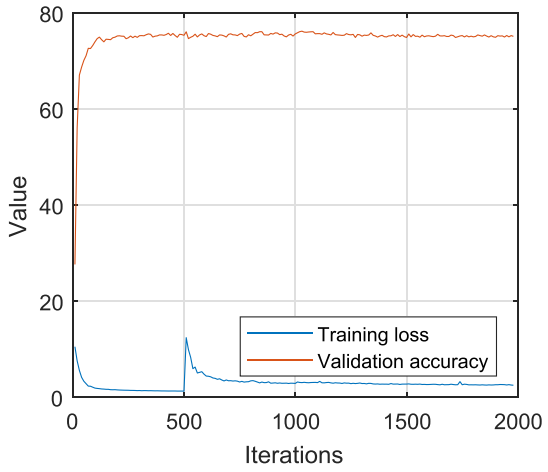


Fig. 17. Training loss and validation accuracy versus iterations for the proposed method. The loss suddenly increases at 500 iterations because the losses $\mathcal{L}_{\mathrm{KT}}^{T}$, $\mathcal{L}_{\mathrm{CE}}^{T}$, and $\mathcal{L}_{D}$ for TD samples are first calculated at this time.

to address the problem that some different objects have similar spectra, and some identical objects have different spectra.

6) When the distance between samples is not mapped to the Hilbert space when computing the triplet loss, the classification accuracies also decrease on the three datasets. This demonstrates that the proposed kernel triplet loss can characterize complex nonlinear relationships between samples to improve classification accuracy.
7) The proposed method uses SD samples for discriminative information learning and uses kernel triplet loss together with cross-entropy loss to train the network, which achieved the best performance in the ablation experiments.

2) Comparison of Weights for Kernel Triplet Loss and Domain Classifier Loss: Fig. 18 shows the classification accuracies versus different weights for kernel triplet loss. We can find that the classification accuracy is the lowest when $\lambda_K = 0$. This shows that when kernel triplet loss is not used, the network is unable to learn good enough discriminative features. As λ_K increases, the classification accuracy gradually increases when $\lambda_K < 5$. However, when λ_K is too big, it cannot improve the performance. We can

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TABLE VII
CONTRIBUTION OF EACH COMPONENT OF THE PROPOSED METHOD

(#) samples per class	dataset	Without EMAP	Without Mapping Layer	Without SD	Without CE Loss	Without KT Loss	Without Hilbert	**CFSL-KT**
5	IP	70.12	75.82	75.62	75.93	71.34	75.87	**76.29**
	UP	72.41	89.83	89.19	88.69	86.04	89.86	**89.92**
	Salinas	91.26	94.38	95.08	95.10	90.89	94.32	**95.13**
	Houston	70.36	82.04	82.21	78.41	80.04	82.14	**82.63**
4	IP	67.35	73.27	73.2	73.03	67.41	73.52	**73.58**
	UP	71.32	88.51	87.66	87.73	85.71	88.70	**89.00**
	Salinas	90.41	92.70	93.67	93.27	89.74	92.72	**93.72**
	Houston	68.75	80.98	78.90	74.57	76.61	79.51	**81.02**
3	IP	62.82	68.31	68.21	67.74	62.17	67.95	**68.44**
	UP	68.71	86.82	85.20	85.74	82.10	86.58	**87.23**
	Salinas	90.02	91.37	91.04	91.86	87.88	91.08	**91.91**
	Houston	62.93	76.56	74.53	69.57	72.08	75.47	**76.60**
2	IP	58.62	61.79	62.04	60.96	54.19	61.36	**62.19**
	UP	60.21	79.81	78.20	78.54	76.27	80.78	**80.80**
	Salinas	89.67	88.97	89.62	89.41	86.14	88.74	**89.89**
	Houston	51.25	68.88	67.91	65.0	65.02	68.56	**69.81**
1	IP	43.34	47.30	45.93	46.48	46.37	46.94	**47.89**
	UP	64.36	67.11	65.11	64.21	64.05	66.54	**67.31**
	Salinas	71.75	75.10	72.59	76.36	75.15	76.46	**76.55**
	Houston	49.28	51.23	51.81	49.58	49.27	52.09	**52.64**
average	IP	60.45	65.30	64.77	64.83	60.29	65.13	**65.68**
	UP	67.40	82.42	80.98	80.98	78.83	82.49	**82.85**
	Salinas	86.62	88.50	89.20	89.20	85.96	88.66	**89.44**
	Houston	60.51	71.94	71.07	67.43	68.60	71.55	**72.54**
all mean		68.75	77.04	76.51	75.61	73.42	76.96	**77.63**

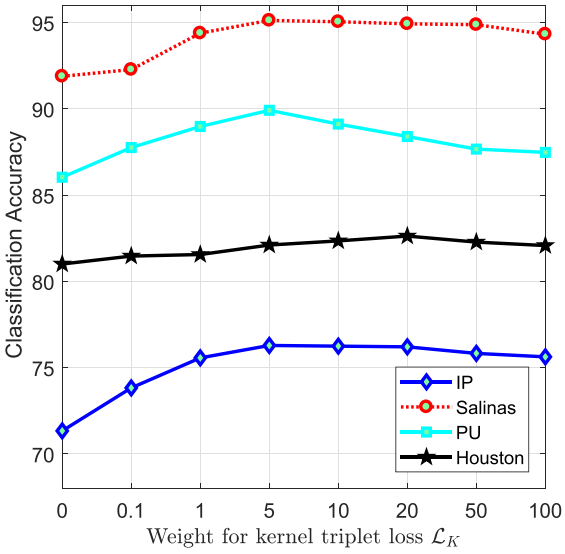


Fig. 18. Comparison of weights for kernel triplet loss.

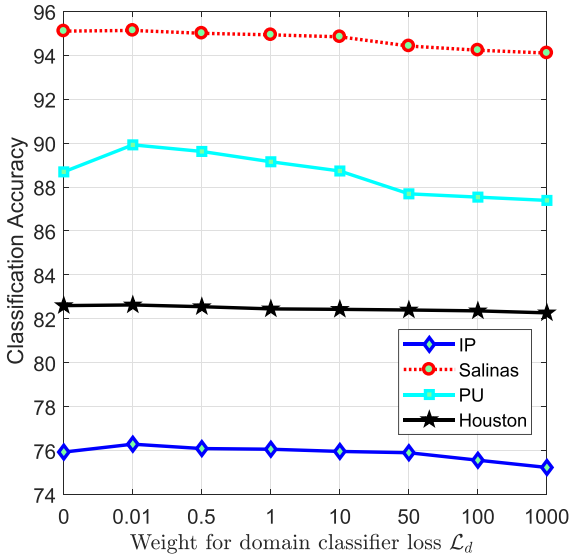


Fig. 19. Comparison of weights for domain classifier loss.

select an appropriate weight between 3 and 10 to get good enough accuracy.

Fig. 19 shows classification accuracies versus weights for domain classifier loss. We can find that the classification accuracy is highest when $\lambda_D = 0.01$. Performance declines when $\lambda_D = 0$ or λ_D increases. Hence, the proposed domain alignment method can improve the accuracy.

In summary, combining the proposed kernel triplet loss and cross-entropy loss can improve accuracy.

V. CONCLUSION

The experiments suggest some conclusions as follows.

1) Compared with traditional CNN-based methods, the proposed CFSL-KT achieved much better performance because the proposed method fully exploited all discriminant information with enough labeled samples from the SD and few labeled samples from the TD.
2) Compared with state-of-the-art cross-domain FSL methods, the proposed CFSL-KT also gets significant

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improvement because the proposed method considers all possible positive and negative pairs in a training batch instead of only considering pairs between the support and query sets, and the proposed kernel triplet loss can map samples to a higher dimensional space for better classification.

In our future work, we will investigate how to better align the source and target data class by class, improve the feature extraction network by the transformer method, and further improve the performance for HSI classification.

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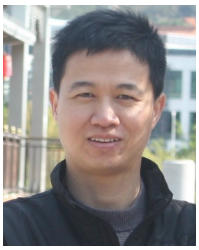
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Ke-Kun Huang (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees in applied mathematics from Sun Yat-sen University, Guangzhou, China, in 2002, 2005, and 2016, respectively.

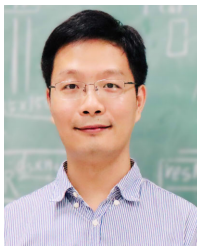
He is currently a Professor with the Department of Mathematics, Jiaying University, Meizhou, China. He has published over 20 papers in top journals, including IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS (TNNLS), IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON CYBERNETICS (TCYB), and *Pattern Recognition*. His research interests include image processing and pattern recognition.

Dr. Huang was elected as a Nanyue Excellent Teacher of Guangdong province.



Hao-Tian Yuan received the M.S. degree from South China Agricultural University, Guangzhou, China, in 2023, where he is currently pursuing the Ph.D. degree with the School of Engineering.

His research interests include hyperspectral imagery processing, pattern recognition, and crop precise classification.



Chuan-Xian Ren (Member, IEEE) received the Ph.D. degree in mathematics from Sun Yat-sen University, Guangzhou, China, in 2010.

From 2010 to 2011, he was a Senior Research Associate with the Department of Electronic Engineering, City University of Hong Kong, Hong Kong. He is currently a Professor with the School of Mathematics, Sun Yat-sen University. His research interests include visual pattern analysis and machine learning.



Yue-En Hou received the B.Sc. and M.Sc. degrees from the Wuhan University of Technology, Wuhan, Hubei, China, in 2005 and 2009, respectively, and the Ph.D. degree from the South China University of Technology, Guangzhou, China, in 2014.

He is currently an Associate Professor with the Computer School, Jiaying University, Meizhou, Guangdong, China. His research interests include pattern recognition and robot control.



Jie-Li Duan received the B.S. degree from Shanxi Agricultural University, Jinzhong, China, in 1997, and the M.S. and Ph.D. degrees from South China Agricultural University, Guangzhou, China, in 2008 and 2018, respectively.

She is currently a Professor with the College of Engineering, South China Agricultural University. Her research interests include agricultural machinery, and agricultural engineering and information.



Zhou Yang received the B.S. and M.S. degrees from Shanxi Agricultural University, Jinzhong, China, in 1995 and 1998, respectively, and the Ph.D. degree from South China Agricultural University, Guangzhou, China, in 2001.

He is currently a Professor with the College of Engineering, South China Agricultural University. He is also a Party Secretary with Guangdong Ocean University, Zhanjiang, China. His research interests include the mechanization of fruit production and intelligent agricultural machinery equipment.

Dr. Yang is also a member of the Chinese Society of Agricultural Engineering.

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